Unknotting number equals genus for the Baker–Kegel and Himeno L-space knots

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Abstract

For an L-space knot $u \geq g_4 = g$. We show $u = g$ for two infinite families of hyperbolic L-space knots that are not braid positive, or not known to be: the knots K_k of Baker and Kegel and the knots K_n of Himeno. Both are members of a two-parameter family $H(n, m)$ of closed $2n$-braids, and $u = g = g_4 = (mn^2 - n + 2)/2$ for every $H(n, m)$, by an explicit unknotting and a braided Seifert surface. The same argument settles the L-space $(2, q)$-cables of $T(2, 2k + 1)$ that are not positive braids. The L-space hypothesis is not superfluous: as Bode and Truöl observe, the mirror of $12n_{642}$ is fibered and strongly quasipositive, with $g = 2$ and $u \geq 3$.

1 Context

An L-space knot is fibered and strongly quasipositive, and $g = g_4 = \tau$ [14, 12, 6]. Since $u \geq g_4$ for every knot, $u \geq g$, and the statement $u = g$ for every L-space knot is the statement $u \leq g$. It was produced as candidate K32 by a program that fits inequalities to the KnotInfo table [10]; the companion note on $\det \leq 1 + |\sigma|$ describes the program. We have not found the statement in the literature; the K3 problem list asks about genus and unknotting number for positive fibered and for alternating fibered knots [8, Problem 1.5], not for L-space knots.

It holds for braid positive knots, where braid relations give an unknotting sequence of length g [1, 15], and Kegel, Lewark, Manikandan, Misev, Mousseau and Silvero show that every positive braid diagram of a genus g knot has g crossings whose change gives the unknot [9]. This covers the torus knots, the $(1, 1)$ L-space knots, which are closures of positive braids [13, Theorem 2], and all L-space knots of the SnapPy census but one [3]. It also holds for strongly quasipositive 3-braid knots [11]. What remains are the L-space knots that are not braid positive. The ones we know are the families of Baker–Kegel [3] and Himeno [7] and some iterated torus knots, such as the $(2, 3)$-cable of the trefoil [3, Example 1].

2 The family

Fix $n \geq 2$. In the braid group B_{2n} let

$$L_k = \sigma_{n-k}\,\sigma_{n-k+2}\cdots\sigma_{n+k} \quad (0 \leq k \leq n-1), \qquad X_n = L_0 L_1 \cdots L_{n-1} L_{n-2} \cdots L_0,$$

$$T_n = \sigma_1^{-1}\cdots\sigma_{n-1}^{-1}\,\sigma_n\sigma_{n-1}\cdots\sigma_1\,\sigma_1\sigma_2\cdots\sigma_n,$$

and for odd $m \geq 1$ let $H(n, m)$ be the closure of $X_n^m T_n$. The letters of each L_k commute. X_n is the n-cable of a positive crossing between two bundles of n strands. The closure is a knot: the permutation of X_n^m exchanges i and $n + i$, that of T_n is an n-cycle on $1, \ldots, n$, and their product is a $2n$-cycle.

Since $X_2 = \sigma_2\sigma_1\sigma_3\sigma_2$ and $T_2 = \sigma_1^{-1}\sigma_2\sigma_1\sigma_1\sigma_2$, the knot $H(2, 2k+1)$ is the knot K_k of Baker and Kegel [3], a hyperbolic L-space knot of genus $4k+2$ for $k \geq 1$, not braid positive for $k = 1$ (K_1 is the census knot o9_30634) and not known to be braid positive for $k \geq 2$. The knot $H(n, 3)$ is the knot K_n of Himeno [7], hyperbolic for $n \geq 2$, and for even n an L-space knot of genus $(3n^2 - n)/2 + 1$ that is not braid positive.

3 The theorem

Theorem 1. *For $n \geq 2$ and odd $m \geq 1$,* $u(H(n,m)) = g(H(n,m)) = g_4(H(n,m)) = (mn^2 - n + 2)/2$.

Corollary 1. *$u = g$ for every Baker–Kegel knot K_k and every Himeno knot K_n. These are L-space knots for all k [3], for even n [7], and for $n = 3, 5$ by knot Floer homology, so for them this is an instance of the statement $u = g$ for L-space knots.*

Proof of Theorem 1. Lower bound. The word $X_n^m T_n$ has $mn^2 + 2n$ positive and $n - 1$ negative letters, so writhe $w = mn^2 + n + 1$, on $2n$ strands. Rudolph's slice–Bennequin inequality [16] gives $g_4 \geq (w - 2n + 1)/2 = (mn^2 - n + 2)/2$, and $u \geq g_4$.

Genus. $T_n = a\,\sigma_1 \cdots \sigma_n$ with $a = (\sigma_n \cdots \sigma_2)\,\sigma_1\,(\sigma_n \cdots \sigma_2)^{-1}$ (Step 4 below), a positive half-twisted band joining the first and the $(n+1)$-st strand that passes on one side of the strands between them. So $X_n^m T_n$ is a product of $mn^2 + n + 1$ positive bands, and its closure bounds the braided surface with one disk per strand and one band per factor [15], a Seifert surface of genus $(mn^2 + n + 1 - 2n + 1)/2 = (mn^2 - n + 2)/2$. Hence $g \leq (mn^2 - n + 2)/2 \leq g_4 \leq g$.

Upper bound. A crossing change in the closed braid replaces a letter $\sigma_i^{\pm 1}$ by $\sigma_i^{\mp 1}$. We make $(mn^2 - n + 2)/2$ of them and reach the unknot. Matching explicit crossing changes against the slice–Bennequin bound is a standard way to compute unknotting numbers of closed braids [17, §4], [9, §2]; what is specific here is the cancellation of the palindromic blocks.

Step 1. Read backwards, X_n is $L_0 \cdots L_{n-2} L_{n-1} L_{n-2} \cdots L_0$ with the letters of each layer permuted, so it represents the same braid; hence the word obtained from X_n by inverting every letter represents X_n^{-1}. Changing all crossings of the second, fourth, …, $(m-1)$-st copy of X_n turns $X_n^m T_n$ into $X_n T_n$, with $n^2(m-1)/2$ changes.

Step 2. Write $X_n = C L_{n-1} \bar{C}$ with $C = L_0 \cdots L_{n-2}$ and $\bar{C} = L_{n-2} \cdots L_0$. Changing the $n(n-1)/2$ crossings of $\bar{C}$ replaces it by C^{-1}.

Step 3. $T_n = a\,\sigma_1 \cdots \sigma_n$ with $a = (\sigma_{n-1} \cdots \sigma_1)^{-1} \sigma_n (\sigma_{n-1} \cdots \sigma_1)$. Changing the crossing σ_n inside a replaces a by a^{-1}. In all, $n^2(m-1)/2 + n(n-1)/2 + 1 = (mn^2 - n + 2)/2$ changes, giving the braid $C L_{n-1} C^{-1} R$ with $R = a^{-1}\sigma_1 \cdots \sigma_n \in B_{n+1}$.

Step 4: the closure of $C L_{n-1} C^{-1} R$ is the unknot. It is the closure of the conjugate $L_{n-1} \cdot C^{-1} R C$. Call the strands at the top of X_n $A_1, \ldots, A_n$ (positions $1, \ldots, n$) and $B_1, \ldots, B_n$ (positions $n+1, \ldots, 2n$). Every crossing of X_n is between an A strand and a B strand, with the B strand over every time in the right-handed drawing of Figure 2. C performs the crossings of A_i with B_j for $j < i$, after which A_i is at position $2i - 1$ and B_i at $2i$, and L_{n-1} crosses A_i with B_i. In the conjugate, after L_{n-1} the strand B_i is at position $2i - 1$ and A_i at $2i$. The word C^{-1} moves the strands at odd positions back to positions $1, \ldots, n$ and those at even positions to $n+1, \ldots, 2n$; since C^{-1} undoes C, the strand coming from an even position passes over every strand it meets, in C^{-1} and again in C. So R acts on $B_1, \ldots, B_n, A_1$ and leaves $A_2, \ldots, A_n$ alone, and each A_i with $i \geq 2$, followed from its crossing in L_{n-1} once around the closed braid back to that crossing, lies over every strand it meets. Lifting that arc above the diagram and shrinking it removes the strand and its crossing. After doing so for $i = n, \ldots, 2$ the closure is that of

$$\sigma_1 \cdot (\sigma_2^{-1} \cdots \sigma_n^{-1}) \cdot R \cdot (\sigma_n \cdots \sigma_2) \in B_{n+1},$$

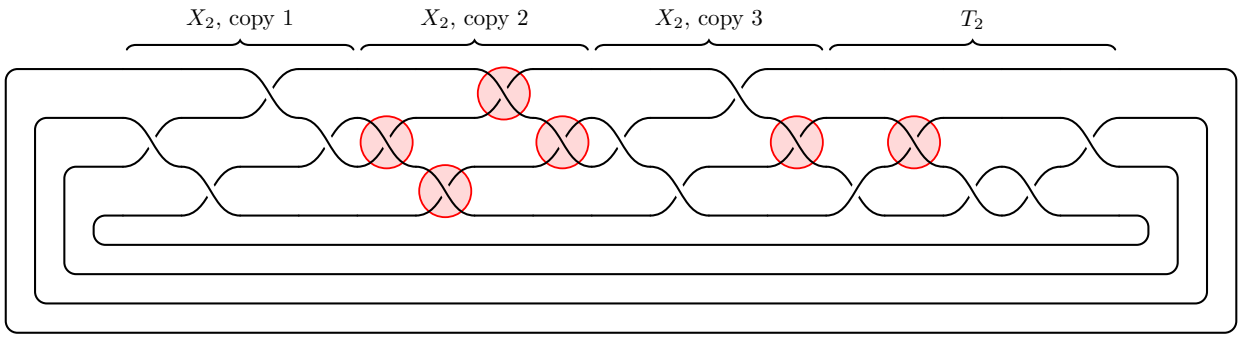


Figure 1: $H(2,3)$, the Baker–Kegel knot K_1 (census o9_30634): the closure of $X_2^3 T_2$ with $X_2 = \sigma_2\sigma_1\sigma_3\sigma_2$ and $T_2 = \sigma_1^{-1}\sigma_2\sigma_1\sigma_1\sigma_2$. The six circled crossings are the changes of Steps 1–3: all of the second copy of X_2, which then cancels the first; the last crossing of the third copy ($\bar{C} = \sigma_2$); and the σ_2 inside a in T_2. Six is g.

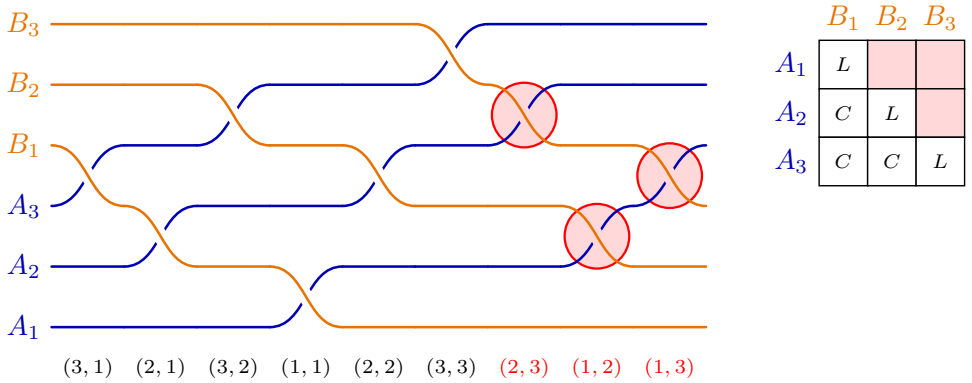


Figure 2: One copy of X_3, with the strands A_i blue and B_j orange and, under each crossing, its pair (i, j). B is over A at every crossing. The circled crossings form $\bar{C}$, the pairs with $j > i$, which Step 2 changes. Right: the same pairs as a grid, C for $j < i$ and L for the middle layer L_2.

where σ_1 is the crossing of A_1 with B_1, the factor before R moves A_1 to position $n+1$ over $B_2, \ldots, B_n$, and the factor after R moves the strand then in position $n+1$ back in the same way. Conjugating by $\sigma_n \cdots \sigma_2$ and using $\sigma_n \cdots \sigma_2\, \sigma_1\, \sigma_2^{-1} \cdots \sigma_n^{-1} = (\sigma_{n-1} \cdots \sigma_1)^{-1} \sigma_n (\sigma_{n-1} \cdots \sigma_1) = a$, which follows by induction from $\sigma_{i+1}\sigma_i\sigma_{i+1}^{-1} = \sigma_i^{-1}\sigma_{i+1}\sigma_i$, the closure is that of $aR = \sigma_1\sigma_2 \cdots \sigma_n \in B_{n+1}$, the unknot. □

Without the change in Step 3 the same reduction gives the closure of $a^2\sigma_1 \cdots \sigma_n$, a knot of genus one (its braided surface has $n+2$ bands on $n+1$ disks); the change in Step 3 unknots it.

Figure 1 shows Steps 1–3 on the smallest case, $H(2,3)$, and Figure 2 shows the triangle of Step 2 for $n = 3$. Both are drawn by `note/figures/make_figures.py` from the same words as the computations, and `check_figures.py` confirms with SnapPy that the drawn $H(2,3)$ is o9_30634 with $\tau = g$, that changing the circled crossings gives the unknot, and that in this drawing each A_i with $i \geq 2$ lies over every strand it meets in Step 4. The braids run from left to right, with position 1 at the bottom and every letter σ_i a right-handed crossing.

4 Cables

Corollary 2. *Every L-space $(2, q)$-cable of $T(2, 2k + 1)$ has $u = g$.*

Proof. The cable is the closure of $X_2^{2k+1}\sigma_1^e$ with $e = q - 2(2k + 1)$, and it is an L-space knot exactly when $q \geq 2(2k - 1)$ [4, 5], that is $e \in \{-3, -1, 1, 3, \dots\}$. For $e \geq 1$ the word is positive. For $e = -1$, Step 1 with k cancelled copies of X_2 ($4k$ changes) leaves $X_2\sigma_1^{-1}$, whose closure is the unknot, and $g = 4k$. For $e = -3$, cancelling $k - 1$ copies ($4k - 4$ changes) leaves $X_2^3\sigma_1^{-3}$, the $(2, 3)$-cable of the trefoil, which three changes unknot (its first three letters); $g = 4k - 1$. □

5 The hypothesis is needed

Proposition 1 (Bode–Truöl [2])**.** *The mirror of $12n_{642}$ is fibered and strongly quasipositive, has genus 2, is not an L-space knot, and has unknotting number at least 3.*

This is [2, §2.7.2 and Example 6.1]; we include the computation as a check.

Proof. Knot Floer homology gives genus 2, fiberedness and $|\tau| = 2$; the chirality with $\tau = g$ is strongly quasipositive, since a fibered knot with $\tau = g$ is [6]. Its total rank is 37, more than $2g + 1$, so it is not an L-space knot. A Seifert matrix V has $\det(V + V^T) = \pm 27$ and $V + V^T$ has nullity 3 over $\mathbb{F}_3$, so the first homology of the double branched cover is $(\mathbb{Z}/3)^3$ and needs three generators; by Wendt [18] this is a lower bound for the unknotting number. □

KnotInfo lists $u(12n_{642}) \in \{3, 4\}$. So $u = g$ fails for fibered strongly quasipositive knots, holds for braid positive ones, and for fibered positive knots is a conjecture of Murasugi and Przytycki [8, Problem 1.5(a)], still open, for which [9] propose a likely counterexample of genus 7.

6 Computations

All in `verification/families`. `unknotting.py`: for $2 \leq n \leq 12$ and $m \in \{1, 3, 5, 7\}$ (44 cases, up to 499 changes) the number of changes equals the slice–Bennequin bound and the changed word closes to the unknot (SnapPy simplification to no crossings, or knot Floer homology of total rank 1). `cables.py`: the two non-positive cases for $1 \leq k \leq 8$ (16 cables), with the genus from knot Floer homology. `fibered_sqp.py`: the proposition. `lspace.py`: which $H(n, m)$ are L-space knots by knot Floer homology; for the 18 members computed, the genus it gives equals $(mn^2 - n + 2)/2$. Tools: SnapPy 3.3.2 with `knot_floer_homology`.

7 Questions

Question 1. *Is $u = g$ known, or expected, for every L-space knot?*

Question 2. *Besides the two families above and some iterated torus knots, which L-space knots are known not to be braid positive?*

References

[1] M. Boileau, C. Weber, *Le problème de J. Milnor sur le nombre gordien des nœuds algébriques*, Enseign. Math. (2) 30 (1984), 173–222.

[2] B. Bode, P. Truöl, *On $\mathcal{T}$-positive links*, arXiv:2605.10502.

[3] K. L. Baker, M. Kegel, *Census L-space knots are braid positive, except for one that is not*, Algebr. Geom. Topol. 24 (2024); arXiv:2203.12013.

[4] M. Hedden, *On knot Floer homology and cabling II*, Int. Math. Res. Not. (2009).

[5] J. Hom, *A note on cabling and L-space surgeries*, Algebr. Geom. Topol. 11 (2011), 219–223.

[6] M. Hedden, *Notions of positivity and the Ozsváth–Szabó concordance invariant*, J. Knot Theory Ramifications 19 (2010), 617–629.

[7] K. Himeno, *Non-braid positive hyperbolic L-space knots*, arXiv:2506.22934.

[8] R. İ. Baykur, R. Kirby, D. Ruberman (eds.), *K3: A New Problem List in Low-Dimensional Topology*, AMS Math. Surveys and Monographs 295 (2026); preliminary copy on R. Kirby's Berkeley web page.

[9] M. Kegel, L. Lewark, N. Manikandan, F. Misev, L. Mousseau, M. Silvero, *On unknotting fibered positive knots and braids*, arXiv:2312.07339.

[10] C. Livingston and A. H. Moore, *KnotInfo: Table of Knot Invariants*, knotinfo.org (accessed 14 September 2026).

[11] E.-K. Lee, S.-J. Lee, *Unknotting number and genus of 3-braid knots*, J. Knot Theory Ramifications 22 (2013), 1350047; arXiv:1305.3455.

[12] Y. Ni, *Knot Floer homology detects fibred knots*, Invent. Math. 170 (2007), 577–608.

[13] Z. Nie, *An explicit description of $(1,1)$ L-space knots, and non-left-orderable surgeries*, arXiv:2102.10891.

[14] P. Ozsváth, Z. Szabó, *On knot Floer homology and lens space surgeries*, Topology 44 (2005), 1281–1300.

[15] L. Rudolph, *Braided surfaces and Seifert ribbons for closed braids*, Comment. Math. Helv. 58 (1983), 1–37.

[16] L. Rudolph, *Quasipositivity as an obstruction to sliceness*, Bull. Amer. Math. Soc. 29 (1993), 51–59.

[17] A. Stoimenow, *Positive knots, closed braids and the Jones polynomial*, Ann. Sc. Norm. Super. Pisa Cl. Sci. (5) 2 (2003), 237–285; arXiv:math/9805078.

[18] H. Wendt, *Die gordische Auflösung von Knoten*, Math. Z. 42 (1937), 680–696.